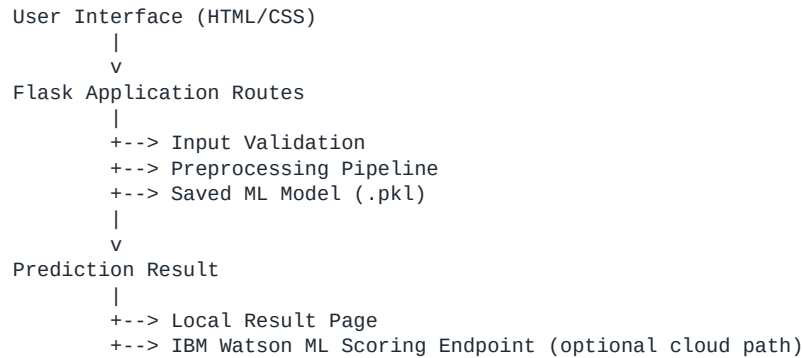


Solution Architecture

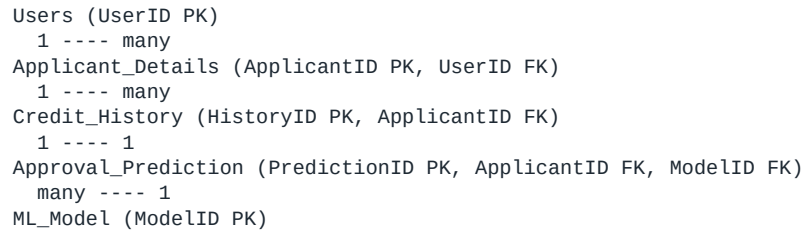
3. Project Design Phase | Credit Card Approval Prediction

Prepared For	Prepared By	Domain	Date
Credit Card Approval Prediction	Ankitha Vana	Finance	July 1, 2026

Architecture Diagram



ER Diagram Description



Entity	Important Attributes
Users	UserID, Name, Email, Password, Role
Applicant_Details	ApplicantID, UserID, IncomeType, EducationType, FamilyStatus, HousingType, EmploymentDays
Credit_History	HistoryID, ApplicantID, MonthBalance, Status, OverdueFlag
ML_Model	ModelID, ModelName, Version, Accuracy, CreatedDate
Approval_Prediction	PredictionID, ApplicantID, ModelID, PredictionLabel, PredictionDate

Deployment Architecture

- Local development happens in Jupyter Notebook and PyCharm or VS Code.
- The trained model is serialized using pickle/joblib.
- The Flask application loads the saved model at startup.
- For cloud deployment, the model asset is uploaded to IBM Watson Machine Learning and exposed through a scoring endpoint.